

Exercises Day 1

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You already made exercises 1 and 2 in the slides. For exercises 3 to 7, you will do them in RStudio. Please make sure that you have R and RStudio installed on your laptop.

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Exercise 1: selection within a vector

- Create a vector named `vec` that consists of the numbers from 11 to 30
- Select the 7th element of the vector
- Select all elements except the 15th. Hint: use a minus sign
- Select the 2nd and 5th element of the vector
- Select only the odd valued elements of `vec`.

```
# a.  
vec <- seq(11,30)  
# b.  
vec[7]
```

```
[1] 17
```

```
# c.  
vec[-15]
```

```
[1] 11 12 13 14 15 16 17 18 19 20 21 22 23 24 26 27 28 29 30
```

```
# d.  
vec[c(2,5)]
```

```
[1] 12 15
```

```
# e.  
index <- seq(1,length(vec),by=2)  
vec[index]
```

```
[1] 11 13 15 17 19 21 23 25 27 29
```

Exercise 2: some calculations

- a. Use the elementary functions $/$, $-$, $\wedge$ and the function `sum` to calculate the mean

$$\bar{x} = \sum_{i=1}^n x_i / n$$

and the standard deviation

$$\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 / (n - 1)}$$

of the fare paid by the titanic passengers.

- b. Verify your answer by using the built-in R functions `mean` and `sd`.

We create the data set that we used in the web based interface.

```
titanic <- data.frame(
  pclass=rep(c("1st", "2nd", "3rd"), c(4, 2, 4)),
  survived=c(1, 0, 0, 0, 1, 1, 0, 0, 0, 0),
  name=c("Allison, Master. Hudson Trevor", "Dulles, Mr. William Crothers", "McCarthy,
  age= c(0.9167, 39, 54, 47, 27, 13, 13, 17, NA, NA),
  fare=c(151.55, 29.7, 51.8625, 34.0208, 13.8583, 19.5, 20.25, 8.6625, 7.75, 23.4))
```

```
# a.
fare <- titanic$fare
MeanFare <- sum(fare)/length(fare)
MeanFare
```

```
[1] 36.05541
```

```
# standard deviation (in three baby steps)
numerator <- sum((fare-MeanFare)^2)
denominator <- (length(fare)-1)
StdFare <- sqrt(numerator/denominator)
StdFare
```

```
[1] 42.63766
```

```
# b.
mean(fare)
```

```
[1] 36.05541
```

```
sd(fare)
```

```
[1] 42.63766
```

RStudio

Some useful Keyboard Shortcuts in RStudio for Windows

Over time it is useful to learn a couple of RStudio shortcuts that you frequently use. You can find an overview of them under the **Help → Keyboard Shortcuts help** menu. More shortcuts and tips can be found on the [Appsilon](#) website.

Note

On the mac, Ctrl → Cmd, Alt → Option, and Enter → Return.

- **Ctrl+Enter**: Run current line
- **Ctrl+Alt+I**: Insert new chunk
- **Alt+-**: Insert assignment <-
- **F1**: Show function help page when cursor is on function name
- **Ctrl+Tab**: Switch between tabs in Source Editor
- **Ctrl+1** / **Ctrl+2**: Move focus to Source/Console
- **Tab** / **Ctrl+Space**: Code suggestion/completion
- **In Console**:
 - Arrows **Up/Down** to navigate earlier commands
 - **Ctrl+Up** pops up a listing of latest commands

Create a folder on your computer for the exercises in this R course. Open RStudio and create a new New Project in that folder. Open a new R Script file via the menu:

File → New File → R Script...

Write the code in that R Script file.

Exercise 3: importing data from Excel and first check

We will work with a dataset that gives the survival status of passengers on the Titanic. We provide the dataset in **Excel** format ([Titanic3.xlsx](#)). See [this link](#) for a description of the variables. Our data set has three additional columns: **dob** (date of birth), **family** (whether the person travelled with family) and **agecat** (age categorized in age ranges).

Download the dataset to the raw data subfolder of your project folder. Import the dataset by clicking on **Import Dataset** from the **Environment** pane. Browse for the Excel file.

The first 50 rows of the data set are shown. Look at the code under **Code Preview**. Assign the R object name **titanic** to the dataset under **Import Options** and observe that the code under **Code Preview** has changed.

Look at the mode of variable age. A problem with the mode for age is that missing values are coded as a character value "NA" in our Excel file; it is not recognized as an R type NA value. In the future, you may encounter other representations of missing values, such as **missing**, **404**, **-99** or **unknown**, which should also be treated as NA. Specify that NA represents missing values under the import options. Observe how the code under

*Code Preview **changed again. Copy the code from** Code Preview**. Click on the “Import” button. Paste the code you have copied earlier to your R Script File.

- a. Inspect the data set in spreadsheet-like format that shows up after importing (if it does not show up, you can click on the `titanic` object in the **Environment** pane). Do the variables make sense? Are there any weird values? Sort the data in ascending order of age by clicking on the **age** in the data set.
- b. Use the function `dim` to find the number of rows (passengers) and columns (variables) in the data set.

```
dim(titanic)
```

```
[1] 1309  17
```

Answer: 1309 rows and 17 columns

- c. Use the `str` function to obtain a summary of the basic characteristics per variable. Have a look at the `survived` column. Does it have the mode that you expect? (You can also use the `mode` function to check the modes.)

```
str(titanic)
```

```
tibble [1,309 x 17] (S3: tbl_df/tbl/data.frame)
 $ pclass   : chr [1:1309] "1st" "1st" "1st" "1st" ...
 $ survived : num [1:1309] 1 1 0 0 0 1 1 0 1 0 ...
 $ name     : chr [1:1309] "Allen, Miss. Elisabeth Walton" "Allison, Master. Hudson Tr
 $ sex      : chr [1:1309] "female" "male" "female" "male" ...
 $ age      : num [1:1309] 29 0.917 2 30 25 ...
 $ sibsp    : num [1:1309] 0 1 1 1 1 0 1 0 2 0 ...
 $ parch    : num [1:1309] 0 2 2 2 2 0 0 0 0 0 ...
 $ ticket   : chr [1:1309] "24160" "113781" "113781" "113781" ...
 $ fare     : num [1:1309] 211 152 152 152 152 ...
 $ cabin    : chr [1:1309] "B5" "C22 C26" "C22 C26" "C22 C26" ...
 $ embarked : chr [1:1309] "Southampton" "Southampton" "Southampton" "Southampton" ...
 $ boat     : chr [1:1309] "2" "11" NA NA ...
 $ body     : num [1:1309] NA NA NA 135 NA NA NA NA NA 22 ...
 $ home_dest: chr [1:1309] "St Louis, MO" "Montreal, PQ / Chesterville, ON" "Montreal,
 $ dob      : num [1:1309] -28019 -17762 -18158 -28384 -26558 ...
 $ family   : chr [1:1309] "no" "yes" "yes" "yes" ...
 $ agecat   : chr [1:1309] "(18,30]" "(0,18]" "(0,18]" "(18,30]" ...
```

```
mode(titanic$survived)
```

```
[1] "numeric"
```

survived is of mode numeric, while one would expect it to be a categorical variable of mode character.

Exercise 4: some columns summarized

- Issue the commands

```
head(titanic, 10)
```

```
# A tibble: 10 x 17
  pclass survived name      sex      age sibsp parch ticket  fare cabin embarked
  <chr>      <dbl> <chr>      <chr>   <dbl> <dbl> <dbl> <chr>  <dbl> <chr> <chr>
1 1st          1 Allen, ~ fema~ 29         0      0 24160  211. B5    Southam~
2 1st          1 Allison~ male   0.917      1      2 113781 152. C22 ~ Southam~
3 1st          0 Allison~ fema~ 2         1      2 113781 152. C22 ~ Southam~
4 1st          0 Allison~ male  30         1      2 113781 152. C22 ~ Southam~
5 1st          0 Allison~ fema~ 25         1      2 113781 152. C22 ~ Southam~
6 1st          1 Anderso~ male  48         0      0 19952   26.5 E12    Southam~
7 1st          1 Andrews~ fema~ 63         1      0 13502   78.0 D7     Southam~
8 1st          0 Andrews~ male  39         0      0 112050    0 A36    Southam~
9 1st          1 Appleto~ fema~ 53         2      0 11769   51.5 C101   Southam~
10 1st          0 Artagav~ male  71         0      0 PC 17~  49.5 <NA> Cherbou~
# i 6 more variables: boat <chr>, body <dbl>, home_dest <chr>, dob <dbl>,
#   family <chr>, agecat <chr>
```

```
tail(titanic, 10)
```

```
# A tibble: 10 x 17
  pclass survived name      sex      age sibsp parch ticket  fare cabin embarked
  <chr>      <dbl> <chr>      <chr>   <dbl> <dbl> <dbl> <chr>  <dbl> <chr> <chr>
1 3rd          0 Yasbeck,~ male   27         1      0 2659   14.5 <NA> Cherbou~
2 3rd          1 Yasbeck,~ fema~ 15         1      0 2659   14.5 <NA> Cherbou~
3 3rd          0 Youseff,~ male  45.5        0      0 2628    7.22 <NA> Cherbou~
```

```

4 3rd      0 Yousif, ~ male    NA      0      0 2647      7.22 <NA> Cherbou~
5 3rd      0 Yousseff~ male   NA      0      0 2627      14.5  <NA> Cherbou~
6 3rd      0 Zabour, ~ fema~  14.5    1      0 2665      14.5  <NA> Cherbou~
7 3rd      0 Zabour, ~ fema~  NA      1      0 2665      14.5  <NA> Cherbou~
8 3rd      0 Zakarian~ male   26.5    0      0 2656      7.22 <NA> Cherbou~
9 3rd      0 Zakarian~ male   27      0      0 2670      7.22 <NA> Cherbou~
10 3rd     0 Zimmerma~ male   29      0      0 315082    7.88 <NA> Southam~
# i 6 more variables: boat <chr>, body <dbl>, home_dest <chr>, dob <dbl>,
#   family <chr>, agecat <chr>

```

What do you observe?

Answer: First and last to rows are shown. Not all columns are shown.

- b. The `readxl` package that is used to import the data stores the data as a `tibble`, which is a special type of format on top of the `data.frame` format to store data in R. Override the `tibble` format using the function `as.data.frame`.

```
titanic <- as.data.frame(titanic)
```

Run the `head` and `tail` functions again. What has changed?

```
head(titanic, 10)
```

```

      pclass survived      name      sex      age sibsp parch
1      1st         1 Allen, Miss. Elisabeth Walton female 29.0000      0      0
2      1st         1 Allison, Master. Hudson Trevor  male  0.9167      1      2
3      1st         0 Allison, Miss. Helen Loraine female  2.0000      1      2
4      1st         0 Allison, Mr. Hudson Joshua Crei  male 30.0000      1      2
5      1st         0 Allison, Mrs. Hudson J C (Bessi female 25.0000      1      2
6      1st         1 Anderson, Mr. Harry      male 48.0000      0      0
7      1st         1 Andrews, Miss. Kornelia Theodos female 63.0000      1      0
8      1st         0 Andrews, Mr. Thomas Jr      male 39.0000      0      0
9      1st         1 Appleton, Mrs. Edward Dale (Cha female 53.0000      2      0
10     1st         0 Artagaveytia, Mr. Ramon      male 71.0000      0      0
      ticket      fare      cabin embarked boat body
1      24160 211.3375      B5 Southampton      2  NA
2     113781 151.5500 C22 C26 Southampton     11  NA
3     113781 151.5500 C22 C26 Southampton <NA>  NA
4     113781 151.5500 C22 C26 Southampton <NA> 135

```

5	113781	151.5500	C22	C26	Southampton	<NA>	NA
6	19952	26.5500		E12	Southampton	3	NA
7	13502	77.9583		D7	Southampton	10	NA
8	112050	0.0000		A36	Southampton	<NA>	NA
9	11769	51.4792		C101	Southampton	D	NA
10	PC 17609	49.5042		<NA>	Cherbourg	<NA>	22

			home_dest	dob	family	agecat
1			St Louis, MO	-28019.25	no	(18,30]
2	Montreal, PQ /	Chesterville, ON	-17761.82		yes	(0,18]
3	Montreal, PQ /	Chesterville, ON	-18157.50		yes	(0,18]
4	Montreal, PQ /	Chesterville, ON	-28384.50		yes	(18,30]
5	Montreal, PQ /	Chesterville, ON	-26558.25		yes	(18,30]
6		New York, NY	-34959.00		no	(40,80]
7		Hudson, NY	-40437.75		yes	(40,80]
8		Belfast, NI	-31671.75		no	(30,40]
9		Bayside, Queens, NY	-36785.25		yes	(40,80]
10		Montevideo, Uruguay	-43359.75		no	(40,80]

```
tail(titanic, 10)
```

	pclass	survived		name	sex	age	sibsp	parch
1300	3rd	0		Yasbeck, Mr. Antoni	male	27.0	1	0
1301	3rd	1		Yasbeck, Mrs. Antoni (Selini Al	female	15.0	1	0
1302	3rd	0		Youseff, Mr. Gerious	male	45.5	0	0
1303	3rd	0		Yousif, Mr. Wazli	male	NA	0	0
1304	3rd	0		Yousseff, Mr. Gerious	male	NA	0	0
1305	3rd	0		Zabour, Miss. Hileni	female	14.5	1	0
1306	3rd	0		Zabour, Miss. Thamine	female	NA	1	0
1307	3rd	0		Zakarian, Mr. Mapriededer	male	26.5	0	0
1308	3rd	0		Zakarian, Mr. Ortin	male	27.0	0	0
1309	3rd	0		Zimmerman, Mr. Leo	male	29.0	0	0

	ticket	fare	cabin	embarked	boat	body	home_dest	dob	family
1300	2659	14.4542	<NA>	Cherbourg	C	NA	<NA>	-27288.75	yes
1301	2659	14.4542	<NA>	Cherbourg	<NA>	NA	<NA>	-22905.75	yes
1302	2628	7.2250	<NA>	Cherbourg	<NA>	312	<NA>	-34045.88	no
1303	2647	7.2250	<NA>	Cherbourg	<NA>	NA	<NA>	NA	no
1304	2627	14.4583	<NA>	Cherbourg	<NA>	NA	<NA>	NA	no
1305	2665	14.4542	<NA>	Cherbourg	<NA>	328	<NA>	-22723.12	yes
1306	2665	14.4542	<NA>	Cherbourg	<NA>	NA	<NA>	NA	yes
1307	2656	7.2250	<NA>	Cherbourg	<NA>	304	<NA>	-27106.12	no

```

1308 2670 7.2250 <NA> Cherbourg <NA> NA <NA> -27288.75 no
1309 315082 7.8750 <NA> Southampton <NA> NA <NA> -28019.25 no
    agecat
1300 (18,30]
1301 (0,18]
1302 (40,80]
1303 <NA>
1304 <NA>
1305 (0,18]
1306 <NA>
1307 (18,30]
1308 (18,30]
1309 (18,30]

```

Answer: All columns are shown.

- c. Summarize the whole dataset using the `summary` function. What do you observe for each of the variables?

```
summary(titanic)
```

```

      pclass      survived      name      sex
Length   :1309  Min.    :0.000  Length   :1309  Length   :1309
N.unique  :    3  1st Qu.:0.000  N.unique :1307  N.unique  :    2
N.blank   :    0  Median :0.000  N.blank  :    0  N.blank   :    0
Min.nchar :    3  Mean    :0.382  Min.nchar:   12  Min.nchar:    4
Max.nchar :    3  3rd Qu.:1.000  Max.nchar:   33  Max.nchar:    6
      Max.    :1.000

      age      sibsp      parch      ticket
Min.   : 0.1667  Min.   :0.0000  Min.   :0.000  Length   :1309
1st Qu.:21.0000  1st Qu.:0.0000  1st Qu.:0.000  N.unique  : 929
Median :28.0000  Median :0.0000  Median :0.000  N.blank   :    0
Mean    :29.8811  Mean    :0.4989  Mean    :0.385  Min.nchar:    3
3rd Qu.:39.0000  3rd Qu.:1.0000  3rd Qu.:0.000  Max.nchar:   18
Max.    :80.0000  Max.    :8.0000  Max.    :9.000
NAs     :263

      fare      cabin      embarked      boat
Min.    : 0.000  Length   :1309  Length   :1309  Length   :1309
1st Qu.: 7.896  N.unique  : 186  N.unique  :    3  N.unique  :   27

```

```

Median : 14.454   N.blank :    0   N.blank :    0   N.blank :    0
Mean   : 33.295   Min.nchar:    1   Min.nchar:    9   Min.nchar:    1
3rd Qu.: 31.275   Max.nchar:   15   Max.nchar:   11   Max.nchar:    7
Max.    :512.329   NAs      :1014   NAs      :    2   NAs      : 823
NAs     :1

      body      home_dest      dob      family
Min.    : 1.0    Length    :1309   Min.    : -46647   Length    :1309
1st Qu.: 72.0    N.unique  : 368    1st Qu.: -31672    N.unique  :    2
Median :155.0    N.blank   :    0    Median : -27654    N.blank   :    0
Mean    :160.8    Min.nchar:    5    Mean    : -28341    Min.nchar:    2
3rd Qu.:256.0    Max.nchar:   31    3rd Qu.: -25097    Max.nchar:    3
Max.    :328.0    NAs       : 564    Max.    : -17488
NAs     :1188                    NAs     :263

      agecat
Length    :1309
N.unique  :    4
N.blank   :    0
Min.nchar:    6
Max.nchar:    7
NAs       : 263

```

Many are of class and mode character, and the `summary` function does not give any information. The `dob` variable has negative values.

- d. Compute the 5%, 25%, 50%, 75% and 95% quantiles, the inter-quartile range and the standard deviation for age and fare. You may have to take a look at the `help` files for the functions `quantile`, `IQR` and `sd`.

```
## some quantiles
quantile(titanic$age, probs=c(0.05,0.25,0.5,0.75,0.95), na.rm=TRUE)
```

```

5% 25% 50% 75% 95%
5  21  28  39  57

```

```
quantile(titanic$fare, probs=c(0.05,0.25,0.5,0.75,0.95), na.rm=TRUE)
```

```

      5%      25%      50%      75%      95%
7.2250  7.8958 14.4542 31.2750 133.6500

```

```
## the IQR (note that this is a single number)
IQR(titanic$age, na.rm=TRUE)
```

```
[1] 18
```

```
IQR(titanic$fare, na.rm=TRUE)
```

```
[1] 23.3792
```

```
## the the standard deviation
sd(titanic$age, na.rm = TRUE)
```

```
[1] 14.4135
```

```
sd(titanic$fare, na.rm = TRUE)
```

```
[1] 51.75867
```

- e. For categorical variables, `summary` is not very informative. Use the `table` function to summarize the variables `pclass`, and `sex`. Also give a two-by-two table for sex and survival status using the same `table` function. Again, have a look at the help file if needed.

```
## basic tables
table(titanic$pclass)
```

```
1st 2nd 3rd
323 277 709
```

```
table(titanic$sex)
```

```
female  male
  466    843
```

```
table(titanic$sex, titanic$survived)
```

```
      0    1
female 127 339
male   682 161
```

```
## argument useNA has no added value because there are no missings in sex and survived
table(titanic$sex, titanic$survived, useNA = "always")
```

```
      0    1 <NA>
female 127 339    0
male   682 161    0
<NA>    0    0    0
```

- f. The function `addmargins` adds the row sums and the column sums to the table of survival status by sex. The function `proportions` tabulates proportions instead of absolute numbers.

```
addmargins(table(titanic$sex, titanic$survived))
```

```
      0    1 Sum
female 127 339 466
male   682 161 843
Sum    809 500 1309
```

```
proportions(table(titanic$sex, titanic$survived))
```

```
      0          1
female 0.09702063 0.25897632
male   0.52100840 0.12299465
```

Use the `proportions` function to compute the fraction (or percentage) of survivors by sex.

```
## basic tables
proportions(table(titanic$sex, titanic$survived), margin = 1)
```

```
      0          1
female 0.2725322 0.7274678
male   0.8090154 0.1909846
```

Exercise 5: work with subsets

- a. Take a look at the name, and the home town/destination of all passengers who were older than 70 years. Use the appropriate selection functions.

```
subset(titanic, age>70)[,c("name", "home_dest")]
```

	name	home_dest
10	Artagaveytia, Mr. Ramon	Montevideo, Uruguay
15	Barkworth, Mr. Algernon Henry W	Hessle, Yorks
62	Cavendish, Mrs. Tyrell William Little Onn	Hall, Staffs
136	Goldschmidt, Mr. George B	New York, NY
728	Connors, Mr. Patrick	<NA>
1236	Svensson, Mr. Johan	<NA>

It is seen that there is one person from Uruguay in this group. Select the single record from that person. Did this person travel with relatives?

```
subset(titanic, name=="Artagaveytia, Mr. Ramon")
```

	pclass	survived	name	sex	age	sibsp	parch	ticket
10	1st	0	Artagaveytia, Mr. Ramon	male	71	0	0	PC 17609

	fare	cabin	embarked	boat	body	home_dest	dob	family
10	49.5042	<NA>	Cherbourg	<NA>	22	Montevideo, Uruguay	-43359.75	no

	agecat
10	(40,80]

Answer: the person did not travel with relatives

- b. Make a table of survivor status by sex, but only for the first class passengers. What do you observe? Compare this with the survival status for the 3rd class passengers.

```
first <- xtabs(~sex+survived, data=titanic, subset=(pclass=="1st"))
first
```

	survived
sex	0 1
female	5 139
male	118 61

```
third <- xtabs(~sex+survived, data=titanic, subset=(pclass=="3rd"))
third
```

```
      survived
sex      0    1
female 110 106
male   418  75
```

Answer: females were much more likely to survive than males, especially among those who travelled first class

Exercise 6: make factor variables

- Make passenger class and sex into factor variables.
- Add a variable `status` to the Titanic data set, which gives the survival status of the passengers as a factor variable with labels “no” and “yes” describing whether individuals survived. Make the value “no” the first level.

```
titanic$status <- factor(titanic$survived, labels=c("no","yes"))
```

- Run the `summary` function again. What change do you observe compared to Exercise 4?

```
summary(titanic)
```

pclass		survived		name		sex	
Length	:1309	Min.	:0.000	Length	:1309	Length	:1309
N.unique	: 3	1st Qu.	:0.000	N.unique	:1307	N.unique	: 2
N.blank	: 0	Median	:0.000	N.blank	: 0	N.blank	: 0
Min.nchar	: 3	Mean	:0.382	Min.nchar	: 12	Min.nchar	: 4
Max.nchar	: 3	3rd Qu.	:1.000	Max.nchar	: 33	Max.nchar	: 6
		Max.	:1.000				

age		sibsp		parch		ticket	
Min.	: 0.1667	Min.	:0.0000	Min.	:0.000	Length	:1309
1st Qu.	:21.0000	1st Qu.	:0.0000	1st Qu.	:0.000	N.unique	: 929
Median	:28.0000	Median	:0.0000	Median	:0.000	N.blank	: 0

```

Mean      :29.8811   Mean      :0.4989   Mean      :0.385   Min.nchar:    3
3rd Qu.:39.0000   3rd Qu.:1.0000   3rd Qu.:0.000   Max.nchar:   18
Max.      :80.0000   Max.      :8.0000   Max.      :9.000
NAs       :263

      fare          cabin          embarked          boat
Min.    : 0.000   Length    :1309   Length    :1309   Length    :1309
1st Qu.: 7.896   N.unique   : 186   N.unique   :    3   N.unique   :   27
Median :14.454   N.blank    :    0   N.blank    :    0   N.blank    :    0
Mean    :33.295   Min.nchar:    1   Min.nchar:    9   Min.nchar:    1
3rd Qu.:31.275   Max.nchar:   15   Max.nchar:   11   Max.nchar:    7
Max.    :512.329   NAs        :1014   NAs        :    2   NAs        :  823
NAs      :1

      body          home_dest          dob          family
Min.    : 1.0   Length    :1309   Min.     :-46647   Length    :1309
1st Qu.:72.0   N.unique   : 368   1st Qu.:-31672   N.unique   :    2
Median :155.0   N.blank    :    0   Median :-27654   N.blank    :    0
Mean    :160.8   Min.nchar:    5   Mean     :-28341   Min.nchar:    2
3rd Qu.:256.0   Max.nchar:   31   3rd Qu.:-25097   Max.nchar:    3
Max.    :328.0   NAs        : 564   Max.     :-17488
NAs     :1188   NAs        :263

      agecat          status
Length    :1309   no :809
N.unique   :    4   yes:500
N.blank    :    0
Min.nchar:    6
Max.nchar:    7
NAs        : 263

```

Answer: Passenger class and sex are summarized properly, and status is summarized as a categorical variable.

Exercise 7: working with dates

- a. The variable `dob` gives the date of birth, using days since January 1st, 1960 (this is the time origin used by Stata). Transform this to a true date value using the function `as.Date` (specify the origin argument).

```
titanic$dob <- as.Date(titanic$dob, origin = "1960/1/1")
```

- b. The default display is “year/month/day”. Make a table of the day of the month that the passengers were born, using the `format` function. What do you observe?

```
table(format(titanic$dob, "%d"))
```

```
13 14 15 16
15 940 86 5
```

Answer: All have a day in the middle of the month

- c. What was the earliest date of birth among all the passengers on the boat. Does this correspond to the age of the oldest person on the date that the ship sank (April 15th, 1912)? Use the functions `min` and `max`.

```
min(titanic$dob, na.rm = TRUE)
```

```
[1] "1832-04-14"
```

```
max(titanic$age, na.rm = TRUE)
```

```
[1] 80
```

```
min(titanic$dob, na.rm = TRUE) + max(titanic$age, na.rm = TRUE) * 365.25
```

```
[1] "1912-04-15"
```

Yes, it does correspond